

Selective Focus: Investigating Semantics Sensitivity in Post-training Quantization for Lane Detection

Yunqian Fan^{1,2}, Xiuying Wei², Ruihao Gong^{2,4}, Yuqing Ma^{3,4}, Xiangguo Zhang² Qi Zhang²

Xianglong Liu^{4*}

¹School of Information Science and Technology, [ShanghaiTech University](#) ²SenseTime Research

³Institute of Artificial Intelligence, [Beihang University](#), Beijing, China

⁴State Key Laboratory of Complex & Critical Software Environment, [Beihang University](#), Beijing, China

fanyq2022@shanghaitech.edu.cn, weixiuying966@gmail.com,

{gongruihao, zhangxianguo, zhangqi3}@sensetime.com, {mayuqing, xliu}@buaa.edu.cn

Abstract

Lane detection (LD) plays a crucial role in enhancing the L2+ capabilities of autonomous driving, capturing widespread attention. The Post-Processing Quantization (PTQ) could facilitate the practical application of LD models, enabling fast speeds and limited memories without labeled data. However, prior PTQ methods do not consider the complex LD outputs that contain physical semantics, such as offsets, locations, etc., and thus cannot be directly applied to LD models. In this paper, we pioneeringly investigate **semantic sensitivity** to post-processing for lane detection with a novel Lane Distortion Score. Moreover, we identify two main factors impacting the LD performance after quantization, namely intra-head sensitivity and inter-head sensitivity, where a small quantization error in specific semantics can cause significant lane distortion. Thus, we propose a Selective Focus framework deployed with Semantic Guided Focus and Sensitivity Aware Selection modules, to incorporate post-processing information into PTQ reconstruction. Based on the observed intra-head sensitivity, Semantic Guided Focus is introduced to prioritize foreground-related semantics using a practical proxy. For inter-head sensitivity, we present Sensitivity Aware Selection, efficiently recognizing influential prediction heads and refining the optimization objectives at runtime. Extensive experiments have been done on a wide variety of models including keypoint-, anchor-, curve-, and segmentation-based ones. Our method produces quantized models in minutes on a single GPU and can achieve 6.4% F1 Score improvement on the CULane dataset.

1 Introduction

Deep neural networks have recently sparked great interest in autonomous driving. As a fundamental component in autonomous driving, lane detection (LD) is fundamental for high-level functions such as lane departure warning, lane departure prevention, etc. Lane detection (Qin, Wang, and Li 2020; Wang et al. 2022b; Tabelini et al. 2021) has garnered significant attention and undergone in-depth research, leading to substantial advancements. However, LD models are often required to run on edge devices within limited sizes,

necessitating quantization and introducing formidable challenges to detection performance.

There are two prevalent quantization techniques: Quantization Aware Training (QAT) e.g. (Choi et al. 2018; Esser et al. 2019; Bhalgat et al. 2020; Jain et al. 2020) and Post-training Quantization (PTQ) e.g. (Hubara et al. 2020; Wu et al. 2020a; Wei et al. 2022; Wang et al. 2022a). Though QAT can often yield promising performance, it requires a longer training duration and the whole labeled dataset, raising computation costs and safety concerns. In contrast, PTQ methods have attracted wide attention from both industry and academia due to their speed and label-free nature. Recently, some PTQ methods (Nagel et al. 2020; Li et al. 2021; Wei et al. 2022) propose to tune the weight by reconstructing the original outputs, bringing better performance.

LD models typically regress semantic outputs with physical meanings such as offsets, locations, and angles, and employ complex post-processing to handle these outputs. Notably, the sensitivity of these semantic outputs to post-processing varies, with certain elements having the potential to induce significant lane deformation even with minor perturbations. Prior PTQ approaches employing direct reconstruction methods on feature maps treat all outputs uniformly, overlooking post-processing information.

In this paper, we first propose the **semantic sensitivity** in lane detection models and introduce a Lane Distortion Score to measure the quantization distortion between the original LD model and the corresponding quantized counterpart. Subsequently, we investigate these sensitivities from two perspectives, namely the intra-head sensitivity and the inter-head sensitivity. Specifically, the intra-head sensitivity highlights the heightened sensitivity of a limited number of foreground (lane) regions to quantization noise during post-processing, while the inter-head sensitivity indicates the varying degrees of sensitivity to the quantization of different semantic heads over time, as shown in Figure 4.

To address the sensitivity problems above, we propose a Selective Focus framework to alleviate the semantic sensitivity in post-training quantization for LD models, enhancing the performance. The proposed framework is deployed with a Semantic Guided Focus module and a Sensitivity Aware Selection module, respectively targeting the intra-head sensitivity and the inter-head sensitivity. First, the Se-

*Corresponding author.

Copyright © 2024, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.